## The EdgeX Foundry architecture

The architecture of EdgeX Foundry is the cornerstone of its success and is thoughtfully designed to maximise technological efficiency. It utilises a modular, microservices approach that ensures adaptability and scalability, much like a city with distinct buildings serving unique purposes. This architectural design streamlines data flow, remains platform-agnostic for seamless integration with diverse technologies, and establishes a secure foundation for data exchange and decision-making at the edge. It is a testament to EdgeX Foundry's commitment to delivering a holistic solution for edge computing, akin to a well-planned city that accommodates the dynamic needs of its inhabitants and their varied requirements.

## Platform agnosticism

EdgeX Foundry wholeheartedly embraces diversity. It doesn't favour one platform over another, ensuring compatibility with a vast array of hardware from different OEMs and ODMs, a range of operating systems, cloud providers, and applications that span direct interfaces to sandboxed environments and containers. This platform agnosticism guarantees effortless integration into your current infrastructure, enabling businesses to adapt and innovate without the need for extensive overhauls.

## Microservices flexibility

At the heart of EdgeX Foundry lies a microservices architecture that serves as a foundation for crafting nimble, scalable, and exceptionally flexible services. This adaptability empowers organisations to tailor their IoT solutions precisely to their needs, avoiding the confinement of rigid, monolithic applications. This approach ensures agility in a market that constantly evolves. It boasts support for containerisation, such as Docker or Snap, promoting application portability and simplifying deployment and orchestration. It facilitates the development of interoperable, plug-and-play edge software applications and value-added services.

## Store and forward capability

EdgeX Foundry offers a crucial feature known as 'store and forward capability'. This capability proves essential in situations where network connectivity is intermittent. It enables data to be temporarily stored at the edge, and when a stable connection is re-established, the data is efficiently forwarded, ensuring seamless operations.

Additionally, EdgeX Foundry optimises data handling by compressing historical data and eliminating redundant information within specific time intervals. This not only reduces the data size but also streamlines the process of collecting data from multiple IoT devices before performing computations for data transmission and storage.

## Intelligence at the edge

EdgeX Foundry's strategy of positioning intelligence at the edge serves as a solution to various critical concerns, including actuation latency, bandwidth constraints, storage limitations, and facilitating remote operations. This approach optimises performance and operational efficiency while also acting as a mini-monitor for IoT devices onsite. Furthermore, this intelligent edge placement allows for the effective utilisation of not only higher CPU compute power but also the efficient utilisation of GPU and AI-related CPU resources.

## Brown and greenfield deployments

EdgeX Foundry's architectural design inherently resolves critical interoperability challenges that arise at the convergence of various directions in a distributed IoT edge architecture. This ensures seamless integration with legacy systems, enabling the collection of data from existing sensors and controllers. Consequently, EdgeX Foundry eliminates the need for costly and disruptive infrastructure overhauls in projects such as smart building optimisation. The real-time processing and analysis of data from brownfield devices at the edge result in enhanced energy efficiency and increased occupant comfort.

Moreover, EdgeX Foundry's capability to accommodate both brownfield and greenfield deployments simplifies the gradual expansion and modernisation of IoT systems while honouring existing infrastructure investments. This adaptability proves to be an invaluable asset in real-world applications, where IoT initiatives often comprise a mixture of legacy and cutting-edge technologies.

## Security and manageability

EdgeX Foundry places a strong emphasis on security, offering the capability to process and store sensitive data locally, thereby significantly reducing the risk of potential security breaches. This approach empowers companies to enhance the protection of their critical edge data and devices. It enables on-premise data processing while allowing selective exposure of specific data and applications to the cloud, ensuring data security and compliance with industry regulations.

EdgeX Foundry doesn't limit its focus to security alone; it also streamlines the management of widely distributed compute nodes and effectively scales down to accommodate highly-constrained devices. This simplifies operational and system management, making the platform more accessible and versatile across a wide range of use cases.

## Support on Ubuntu Core, an IoT-optimised Linux distribution

Ubuntu Core is purpose-built to address the distinctive requirements of IoT deployments. Its lightweight design and minimal resource footprint are well-suited for edge devices with constrained